

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Machine Learning

ASSESSMENT TOOL 1 – Analytical Problem Solving (5 QUESTIONS)

Course Code:	MLA0407	Course Title:	Deep Learning
CO Assessed:	CO1	Assessment:	Assessment Tool 1 – Analytical problem solving
Weightage:	35% of CIA	Total Marks:	(5 Questions × 10 Marks)
CO1 Statement:	<i>Apply the fundamental concepts of machine learning and neural networks to design and analyze learning algorithms using perceptrons, multilayer perceptrons, forward propagation, backpropagation, gradient descent optimization techniques, and Maximum Likelihood Estimation for solving real-world problems.</i>		

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Q1. Design a Machine Learning Solution for Predicting Customer Loan Approval

Machine Learning can be used to automate the process of loan approval by analyzing customer information and predicting whether a loan should be approved or rejected. A supervised learning algorithm such as Decision Tree, Logistic Regression, or Random Forest can be used because historical data containing approved and rejected loan applications is available.

Major Stages of Model Development

1. Data Collection

The first step is to collect historical customer loan data. The dataset may contain attributes such as:

- Customer Age
- Annual Income
- Employment Status
- Credit Score
- Existing Loans
- Loan Amount Requested
- Loan Approval Status (Approved/Rejected)

2. Data Preparation

Data preprocessing is performed to improve data quality. This includes:

- Removing duplicate records
- Handling missing values
- Converting categorical values into numerical values
- Normalizing numerical features
- Selecting important features for prediction

3. Splitting the Dataset

The dataset is divided into:

- Training Set (80%) for model training
- Testing Set (20%) for performance evaluation

4. Model Training

A machine learning algorithm such as Decision Tree is trained using the training dataset. During training, the model learns relationships between customer attributes and loan approval decisions.

5. Prediction

After training, the model predicts whether a new customer's loan application should be approved or rejected based on the provided input features.

6. Performance Evaluation

The effectiveness of the model is measured using:

- Accuracy
- Precision
- Recall
- F1-Score
- Confusion Matrix

Workflow of Loan Approval Prediction

Data Collection → Data Preprocessing → Feature Selection → Model Training → Prediction
→ Performance Evaluation

Conclusion

Machine learning provides a reliable and efficient solution for loan approval prediction. By using historical customer data and appropriate learning algorithms, financial institutions can make faster and more accurate lending decisions while reducing risks.

Q2. Architecture of an Artificial Neural Network (ANN)

An Artificial Neural Network (ANN) is a computational model inspired by the structure and functioning of the human brain. It consists of interconnected artificial neurons organized into layers. ANN is widely used for classification, prediction, and pattern recognition tasks.

Structure of ANN

An ANN generally consists of three main layers:

1. Input Layer

The input layer receives data from the external environment. Each neuron in the input layer represents one input feature.

Example:

- Age
- Income
- Credit Score

2. Hidden Layer

The hidden layer performs computations and extracts patterns from the input data. There can be one or more hidden layers in an ANN.

3. Output Layer

The output layer produces the final result of the network. For example, in a loan approval system, the output may indicate whether the loan is approved or rejected.

Components of ANN

Artificial Neuron

A neuron receives inputs, multiplies them by weights, adds a bias, and applies an activation function to generate an output.

Weights (W)

Weights represent the importance of each input feature. Higher weights indicate greater influence on the output.

Bias (b)

Bias helps adjust the output independently of the input values. It improves the flexibility of the model.

Activation Function

Activation functions introduce non-linearity into the network. Common activation functions include:

- Sigmoid
- ReLU
- Tanh

Mathematical Representation

The weighted sum of inputs is calculated as:

$$z = (w_1x_1 + w_2x_2 + \dots + w_nx_n) + b$$

The activation function then produces:

$$y = f(z)$$

Working of ANN

1. Input data is provided to the input layer.
2. Inputs are multiplied by weights.
3. Bias is added to the weighted sum.
4. The activation function processes the result.
5. The output is passed to the next layer.
6. The final output layer generates predictions.

Conclusion

Artificial Neural Networks are powerful machine learning models capable of learning complex patterns from data. The combination of neurons, weights, biases, and activation functions enables ANNs to solve a wide range of prediction and classification problems.

Q3. Working of the Perceptron Learning Algorithm

The Perceptron Learning Algorithm is one of the earliest neural network models used for binary classification problems. It classifies input data into one of two categories by learning appropriate weights and bias values.

Steps of the Perceptron Algorithm

1. Initialize weights and bias.
2. Provide training inputs and target outputs.
3. Compute the weighted sum.
4. Apply the activation function.
5. Calculate the error.
6. Update weights and bias.
7. Repeat until convergence.

Numerical Example

Given:

- Input $x_1 = 1$
- Input $x_2 = 0$
- Target Output (t) = 1
- Initial Weight $w_1 = 0$
- Initial Weight $w_2 = 0$
- Bias $b = 0$
- Learning Rate $\eta = 1$

Step 1: Calculate Net Input

$$\begin{aligned} Net &= (w_1 x_1) + (w_2 x_2) + b \\ Net &= (0 \times 1) + (0 \times 0) + 0 \\ Net &= 0 \end{aligned}$$

Step 2: Apply Activation Function

Since $Net = 0$,

Output $y = 0$

Step 3: Calculate Error

$$Error = Target - Output$$

$$Error = 1 - 0 = 1$$

Step 4: Update Weight w_1

$$w_1(new) = w_1(old) + \eta(Error)x_1$$

$$w_1(new) = 0 + 1(1)(1)$$

$$w_1(new) = 1$$

Step 5: Update Weight w_2

$$w_2(new) = 0 + 1(1)(0)$$

$$w_2(new) = 0$$

Step 6: Update Bias

$$b(new) = b(old) + \eta(Error)$$

$$b(new) = 0 + 1(1)$$

$$b(new) = 1$$

Updated Parameters

- $w_1 = 1$
- $w_2 = 0$
- $b = 1$

Conclusion

The Perceptron Learning Algorithm updates weights and bias whenever a classification error occurs. Through repeated iterations, it learns a decision boundary that separates two classes effectively.

Q4. Forward Propagation in a Multilayer Perceptron (MLP)

Forward propagation is the process by which input data moves through the neural network from the input layer to the output layer. During this process, weighted sums and activation functions are applied to generate predictions.

Given Data

Inputs:

- $x_1 = 1$
- $x_2 = 2$

Weights:

- $w_1 = 0.5$
- $w_2 = 0.4$

Bias:

- $b = 0.1$

Activation Function:

- Sigmoid Function

Step 1: Calculate Weighted Sum

$$\begin{aligned}z &= (x_1 w_1) + (x_2 w_2) + b \\z &= (1 \times 0.5) + (2 \times 0.4) + 0.1 \\z &= 0.5 + 0.8 + 0.1 \\z &= 1.4\end{aligned}$$

Step 2: Apply Sigmoid Activation Function

$$\begin{aligned}\sigma(z) &= \frac{1}{1 + e^{-z}} \\ \sigma(1.4) &= \frac{1}{1 + e^{-1.4}} \\ \sigma(1.4) &\approx 0.802\end{aligned}$$

Step 3: Compute Final Output

$$\text{Output} = 0.802$$

Forward Propagation Process

1. Inputs are supplied to the network.
2. Weighted sums are computed.
3. Bias values are added.
4. Activation functions transform the results.
5. Output values are generated.

Importance of Forward Propagation

- Generates predictions.
- Forms the basis for training.
- Helps compute error during learning.

Conclusion

Forward propagation enables a neural network to process input data and generate meaningful predictions. It is a fundamental step in the operation of Multilayer Perceptrons and deep learning models.

Q5. Backpropagation Algorithm Using Gradient Descent

Backpropagation is a supervised learning algorithm used to train neural networks. It minimizes prediction errors by propagating the output error backward through the network and updating weights using Gradient Descent.

Steps of Backpropagation

1. Perform forward propagation.
2. Calculate output error.
3. Propagate error backward.
4. Compute gradients.
5. Update weights and biases.
6. Repeat until the error is minimized.

Numerical Example

Given:

- Target Output = 1
- Actual Output = 0.8
- Learning Rate (η) = 0.1

- Initial Weight = 0.5
- Input = 1

Step 1: Calculate Error

$$Error = Target - Actual\ Output$$

$$Error = 1 - 0.8$$

$$Error = 0.2$$

Step 2: Compute Gradient

$$Gradient = Error \times Input$$

$$Gradient = 0.2 \times 1$$

$$Gradient = 0.2$$

Step 3: Update Weight

$$W_{new} = W_{old} + \eta(Gradient)$$

$$W_{new} = 0.5 + 0.1(0.2)$$

$$W_{new} = 0.5 + 0.02$$

$$W_{new} = 0.52$$

Step 4: Error Propagation

The calculated error is propagated backward from the output layer to hidden layers. Each neuron receives a portion of the error based on its contribution to the final output.

Step 5: Weight Adjustment

Weights connected to neurons are adjusted in the direction that reduces the overall network error.

Role of Gradient Descent

Gradient Descent is an optimization algorithm used to minimize the loss function by adjusting weights iteratively. It moves the weights in the direction of the steepest decrease in error.

Advantages of Backpropagation

- Efficient learning algorithm
- Reduces prediction error
- Supports deep neural networks
- Improves model accuracy

Conclusion

Backpropagation combined with Gradient Descent is the foundation of modern neural network training. By continuously updating weights and biases based on error calculations, the network learns to make increasingly accurate predictions.